IMPERIAL

Detection and photometric analysis of LEO satellite streaks in archival **ALPACA** all-sky optical data

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Project page, code, paper (in prep) & contact details

1. Satellite Contamination

- Dramatic increase in Low Earth Orbit (LEO) satellites since the launch of Starlink in 2019 **(>15,000 now operational**) [1].
- Projections indicate large LEO expansion with **>2M proposed**.
- Significant **light pollution** from both the satellites themselves and their launch craft/debris.
- **IAU-CPS** coordinates efforts to mitigate the impacts of satellite interference on the night sky.

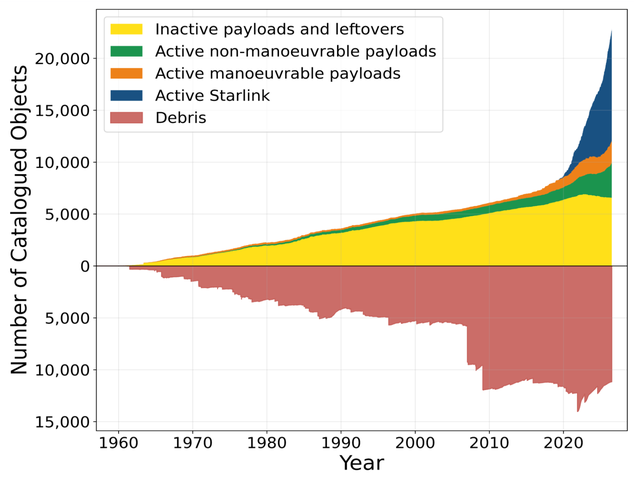
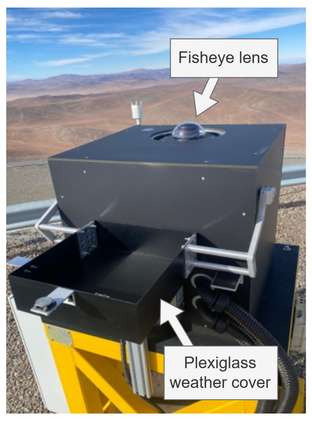


Fig. 1. Increase of LEO satellites over time. Data from Jonathan McDowell [1]



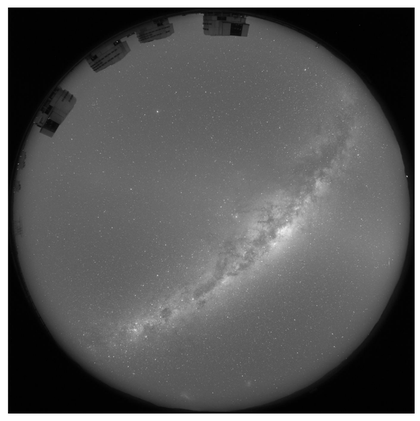


Fig. 2. Left: The ALPACA camera housed at Paranal. Right: An example FITS exposure.

- **ALPACA** is an all-sky weather monitor housed at Paranal using a 12mm lens and a **Sony IMX461 CMOS** sensor.
- Takes continuous 120s monochromatic exposures during nights in the **400-700** nm band.
- Limiting stellar magnitude **~10 mags**

Aim

To quantify the brightness of satellite constellations and catalogued space debris using all-sky (180° FOV) data.

2. Image Processing and Streak Detection

- Astrometric calibration facilitated using **SkyFit2** software [2].
- **Inverse Gnomonic projection** using the pixels RA/DEC location (via SkyFit2 platepar), sampled with **bicubic interpolation**.

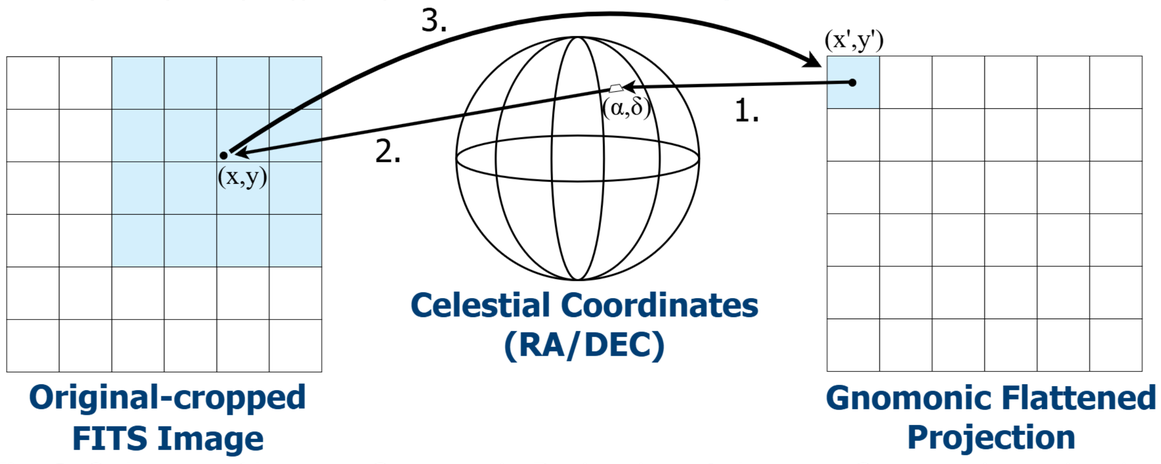


Fig. 3. Schema of Inverse Gnomonic Projection: **1.** Inverse Gnomonic projection, **2.** Pixel locations via SkyFit2, **3.** Bicubic interpolation to obtain Gnomonic Projection pixel value.

- **Progressive Probabilistic Hough** transformations applied onto the gnomonic space, extracting linear streaks. **Glints** detected using line fitted contours.
- IAU CPS **SatChecker** API used to identify potential passes with a **KDTree** finding the closest match [3].

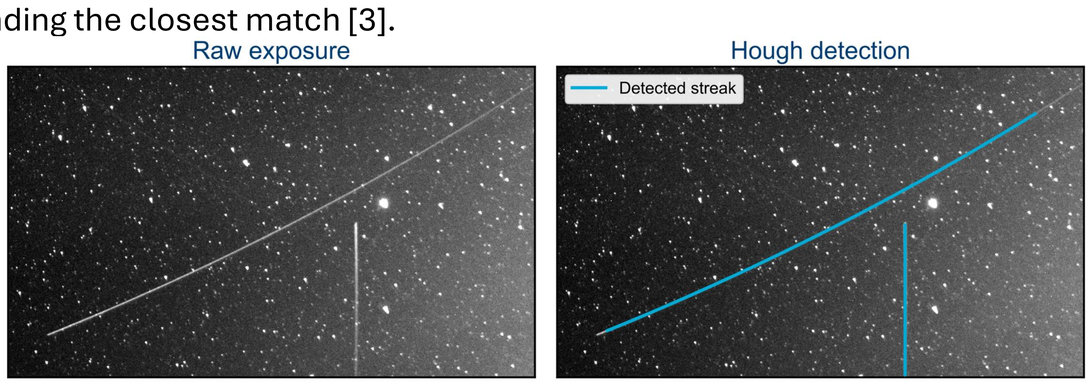


Fig. 4. Example output for the streak detection stage.

3. Brightness determination / validation

- 10 Local stars used to determine magnitude **zero point**.
- **Aperture photometry** applied on 20-pixel segments of extracted streak Hough splines. **Gaussian fitting** and **Hampel** filters applied to improve SNR and mask contaminated segments.
- **TLE-derived** angular rate (SatChecker) converts streak position into elapsed time, enabling streak and per-segment time-resolved magnitudes.

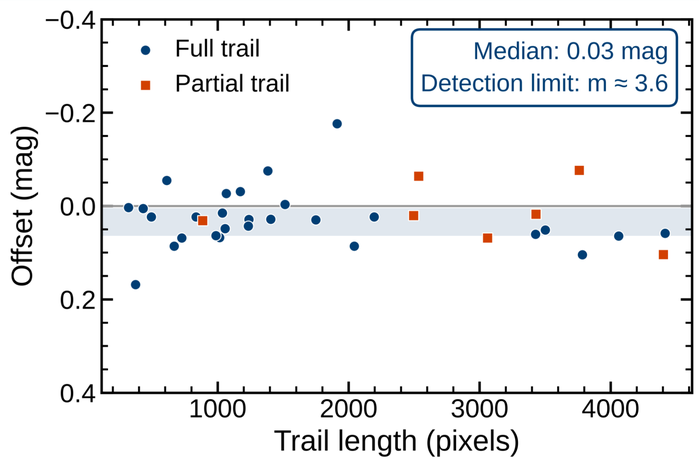


Fig. 5. Photometric offset of simulated streaks injected into real exposures.

- Synthetic streaks are built by integrating a radially interpolated **empirical PSF** with a line segment of random length, orientation and position.
- Streaks are injected into real ALPACA exposures and run through the full pipeline. Recovered flux is compared with injected flux.
- A streak at **3.6 mags** has a 50% chance of detection.

4. Current findings

- **Identified** 300 individual catalogued objects from 584 streaks detected during new moon evenings Feb 2025 – July 2026.
- Faintest detected satellite at **3.8 mags**, agreeing with simulations.
- ALPACA is well suited to analysing **glint** events, even those < 0 mag.

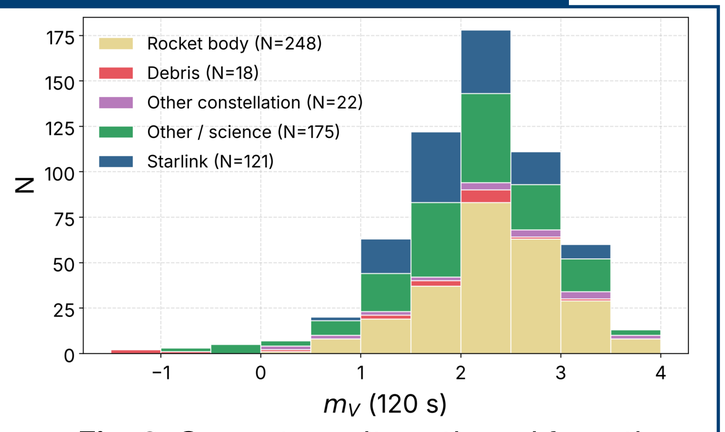


Fig. 6. Current results gathered from the pipeline (new-moon twilights, Feb 2025 – Jul 2026). Rocket parts most common.

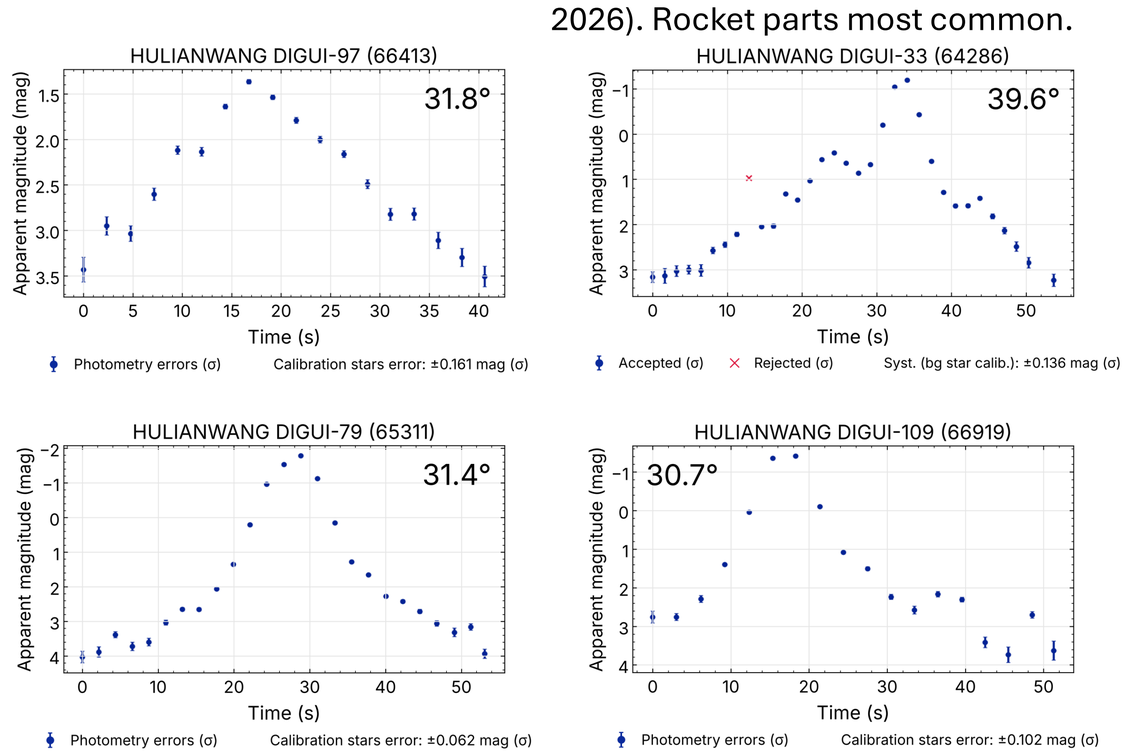


Fig. 7. Brightness plots of 4 selected **Hulianwang** Glints detected Feb 2025 – Jul 2026 (new moon twilight exposures).

Summary

- Developed a pipeline to identify and photometrically quantify satellite streaks using all-sky optical data.
- Produced light curves to analyse changes in brightness.
- Validated outputs using known brightnesses and simulated streaks.
- **Paper in preparation: Bergstrom, O et al.**

5. Further work

- Improved signal to noise detection, requiring improvements to preprocessing and more sensitive line detection algorithms.
- Test pipeline on other all-sky survey cameras, potential to result in ready-to-deploy software.

Acknowledgements

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References

[1] McDowell, Jonathan (2026). Satellite Constellations. https://planet4589.org/space/con/conlist.html. Accessed: 2026-08-11
[2] Vida, D., Šegon, D., Gural, P.S., et al. (2021). The Global Meteor Network – Methodology and first results *MNRAS*, 506, 5046 (2021).
[3] IAU CPS SatHub. SatChecker: satellite position prediction API. https://satchecker.cps.iau.org